Tucker Trivette

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NC State University, Raleigh, NC Expected Graduation: 5/28
GPA: 3.99/4.00, BS in Aerospace Engineering (BSAE)

WORK EXPERIENCE

Astranis Space Technologies — August 2026 - December 2026
Production Quality Engineering Intern — *San Francisco, California*

- Designing and mass-producing **additive-manufactured** signal board covers for in-sourced **conformal coatings.**
- Working with the manufacturing team to **in-source PCBA** late-stage development and testing methods.
- Supporting **satellite hardware** quality assurance processes and communicating with design and manufacturing teams to pinpoint systemic issues and introduce solutions.
- Utilizing **Claude Code** to add features to an inspection auditing tool and fully develop an image documentation interface.

NASA Marshall Space Flight Center — June 2026 - August 2026
Flat Floor Robotics Design and Test Engineering Intern — *Huntsville, Alabama*

- **Designed and integrated a flight-hardware avionics box** in Creo Parametric CAD, with hardware scheduled for microgravity laser welding research testing on a parabolic flight in 2027. Managed **selection and integration** of cRIO controllers, Xiphos electronics board, DC converters, and terminal blocks into a **flight-ready enclosure**. Routed cables and connections within the assembly.
- Developed ground-support components including **displacement laser mounts, laser caps, and weld plates** to enable ground testing and validate performance. Developed a **MATLAB script** and work instructions that enabled the team to run **three load cells simultaneously**, a new capability for the team, allowing **eight in-house CG measurements in approximately 25% of the time** required by the existing method.
- Supported thermal vacuum chamber robotic arm welding testing by **refining CAD collision models for MoveIt Pro**, assisting with chamber setup and operational adjustments, and coordinating with an interdisciplinary team throughout the testing period. Developed a CAD adjustment technique to correct the collision model, **saving approximately one hour after each hardware adjustment.**

VCU Mechanical Engineering — May 2025 - August 2025
Superhard Materials Lab Assistant — *Richmond, Virginia*

- **Led end-to-end materials testing**: sample procurement, hardness indentation testing, and ratio documentation.
- **Fabricated ~40 samples** through arc melting, precision cutting, and polishing for mechanical and structural testing.
- Performed hardness estimation via indentation techniques, **recording and analyzing** indentation measurements.
- **Employed ImageJ** for calculating material ratios, analyzing data, and viewing visual representations of results.

OTHER EXPERIENCE

FSAE Pack Motorsports — August 2025 - Present
Aerodynamics Sub-Team Member — *Raleigh, North Carolina*

- **2027 Season- Designed initial side diffuser model** for 2027 EV car in SolidWorks.
- Responsible for **side diffuser/ side pod end-to-end engineering** development for 2027 EV car.
- **2026 Season-** Designed end wing prototype utilizing **SolidWorks and ANSYS Fluent** for EV car aerodynamic package; **achieved 10.07 L/D Ratio and 34.81 pounds of downforce** in fluid simulation.
- Fabricated past aero package utilizing wet layups, molds, bonding, and clearcoat finish prep.
- Ran ANSYS Fluent Simulations on different element pitches for performance evaluation.

PROJECTS

Aerodynamic Lift/Drag Wind Tunnel — August 2025 – May 2026
Personal Project for FSAE - Hardware Images Featured in LinkedIn Project Section

- **Validated 2026 front wing concept design** with a scaled-down 3D-printed model- **achieved within 1.5% drag error** when compared to ANSYS Fluent simulation results.
- Designed and 3D-printed full structure and test components using **SolidWorks** and additive manufacturing techniques.
- Designed and Machined aluminum structural elements with precision tools, including bandsaws and milling machines.
- Prototyped and integrated sensors for real-time aerodynamic analysis: servo motor (adjustable angle of attack), pressure sensor (wind speed), adjustable test area input, and load cell amplifier (lift and drag measurements).
- **Developed live data visualization and computation modules** for **lift and drag values.**

SKILLS

Modeling: Creo Parametric, SolidWorks **Hardware:** 3D printing, Machine Milling **Simulation:** FEA, ANSYS Fluent, ANSYS Discovery
Software/Data Analysis: Claude Code, MATLAB, Python, SQL, KiCad **Materials:** Microscopic Manipulation, Laser Drilling, Mass Spectroscopy, Arc melting, LabVIEW